

House Price Prediction in King County, Washington

Amin Entezari Behtash Kiani

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Abstract

This paper presents a complete machine learning analysis of the King County House Sales dataset (21,612 residential transactions, 2014–2015), implemented in KNIME Analytics Platform. We compare four regression models — Linear Regression (LR), Decision Tree (DT), Random Forest (RF), and Gradient Boosted Trees (GBT) — under a unified preprocessing pipeline including feature engineering, Z-score normalisation, and log transformation of the target. Random Forest achieves the best performance (Test $R^2=0.703$, RMSE=0.292), confirmed by 5-fold cross-validation (CV RMSE=0.286). Dimensionality reduction via PCA with 12 components (95% retained variance) yields $R^2=0.654$ via LR on the compressed representation. K-Means clustering ($K=2$, Silhouette=0.248) identifies two geographically distinct market segments. Residual analysis confirms unbiased, homoscedastic errors. Feature importance reveals that living space, construction grade, and bathroom count are the primary price determinants.

Keywords: House Price Prediction, Random Forest, Gradient Boosted Trees, PCA, K-Means Clustering, KNIME, King County.

Introduction

HOUSE price prediction is a fundamental problem in real estate economics, with direct implications for buyers, sellers, mortgage lenders, and urban planners. Traditional appraisal methods rely on manual comparison with similar properties, making them inherently subjective, labour-intensive, and difficult to scale. Machine learning offers a data-driven, reproducible alternative that can exploit large transaction datasets to learn complex non-linear relationships between property attributes and sale prices.

This study analyses the King County House Sales dataset from Kaggle [1], which records 21,612 residential property transactions in King County, Washington State (USA) between May 2014 and May 2015. We address the following research question: *which structural and locational features most strongly determine house prices in King County, and how accurately can machine learning models predict them?*

Three alternative approaches are documented in the literature. Hedonic regression [4] models price as a linear function of property attributes but cannot capture non-linearities. Spatial econometrics explicitly models geographic autocorrelation but requires specialised software and parametric assumptions. Deep learning achieves strong performance on large datasets but sacrifices interpretability and requires significantly more data. Our approach differs from these by combining a rigorous comparison of four models of increasing complexity, PCA-based dimensionality reduction, and unsupervised market segmentation — all within a single, reproducible KNIME workflow.

Our main contributions are:

1. A standardised benchmark of four regression models (LR, DT, RF, GBT) under identical preprocessing, holdout, and cross-validation protocols.
2. A PCA analysis that precisely quantifies the information cost of compressing 14 features into 12 principal components.
3. K-Means market segmentation revealing a geographically coherent premium/standard price split.
4. Residual and feature importance analysis validating modelling assumptions and identifying primary price drivers.

The remainder of this paper is organised as follows. Section 2 describes the dataset and preprocessing pipeline.

Section 3 presents the methods. Section 4 reports and discusses all results. Section 5 concludes with limitations and future directions.

Dataset

Dataset Description

The King County House Sales dataset [1] contains 21,613 rows and 21 columns. Each row represents a single residential property transaction. The target variable is **price**, a continuous variable ranging from \$75,000 to \$7,700,000. The raw feature set includes physical attributes (bedrooms, bathrooms, square footage, floors, condition, grade), locational attributes (latitude, longitude, zipcode), and temporal attributes (year built, year renovated). Table 1 summarises the key features used in modelling after preprocessing.

Table 1: Feature set after preprocessing (14 variables).

Feature	Type	Description
bedrooms	Numeric	Number of bedrooms
bathrooms	Numeric	Number of bathrooms
sqft_living	Numeric	Interior living area (sqft)
sqft_lot	Numeric	Lot area (sqft)
floors	Numeric	Number of floors
waterfront	Binary	Waterfront property
view	Numeric	View quality score (0–4)
condition	Numeric	Property condition (1–5)
grade	Numeric	Construction quality (1–13)
sqft_living15	Numeric	Avg neighbour living area
sqft_lot15	Numeric	Avg neighbour lot area
house_age	Numeric	Age at time of sale
renovated	Binary	Has been renovated
basement	Binary	Has a basement

Dataset Limitations

The dataset presents several limitations. It covers a single geographic region (King County) and a single year (2014–2015), which restricts temporal and spatial generalisability. The data do not include macroeconomic variables such as interest rates or housing supply, which are known price drivers. Furthermore, the dataset does not capture subjective value factors such as interior quality, neighbourhood safety, or school district ratings, which are typically important determinants of residential prices.

Preprocessing

Outlier Removal and Column Cleaning

One clear data entry error is identified and removed: a property listed with 33 bedrooms in only 1,620 sqft of living space. After this removal, 21,612 records remain. Non-informative columns (**id**, **date**, **zipcode**) are

removed, as they carry no predictive signal. `zipcode` is excluded because geographic information is already captured by `lat` and `long` in the clustering branch.

Feature Engineering

Three new features are derived from raw columns to improve predictive signal:

- `house_age` = `2015 - yr_built`: the age of the property at time of sale, which is more informative than the raw construction year.
- `renovated` = 1 if `yr_renovated` > 0, else 0: a binary indicator capturing whether the property has been upgraded, which is more interpretable than the raw year.
- `basement` = 1 if `sqft_basement` > 0, else 0: presence of a basement is a cleaner signal than raw square footage for this attribute.

After creating these features, the original columns `yr_built`, `yr_renovated`, `sqft_basement`, and `sqft_above` are removed. `sqft_above` is redundant with `sqft_living` (correlation $\rho = 0.88$) and is removed to reduce multicollinearity.

Log Transformation of Target

The raw `price` variable is strongly right-skewed, with a small number of very high-value properties distorting the distribution. We apply a log transformation: `log_price` = $\ln(\text{price})$. This choice is motivated by two considerations. First, price relationships in real estate are multiplicative rather than additive — a 10% increase in living space adds proportionally more value to a large luxury property than to a small apartment. Second, the log-normal distribution of prices better satisfies the normality assumption of linear regression. Figure 1 confirms that log transformation produces an approximately Gaussian distribution. All model targets and evaluation metrics refer to the log-price scale.

Normalisation and Train/Test Split

Z-score normalisation (mean=0, std=1) is applied to all 14 features. Crucially, the normalisation parameters (mean and standard deviation) are computed exclusively on the training set and then applied to the test set via a Normalizer Apply node. This design prevents data leakage: the model never has access to information from the test set during training. The dataset is split 70/30 using random stratified sampling with seed=42, yielding 15,128 training records and 6,484 test records.

Correlation Analysis

Table 2 reports the strongest pairwise Pearson correlations among the 14 features. `sqft_living` and `grade` are the features most strongly correlated with `log_price`. The high mutual correlations among size-related features (`sqft_living`, `sqft_living15`, `bathrooms`: $\rho > 0.75$) indicate partial redundancy within this group, which motivated the PCA analysis described in Section 4.2. Negative correlations between `house_age` and both `floors` ($\rho = -0.489$) and `grade` ($\rho = -0.447$) confirm that newer properties tend to be taller and built to higher quality standards.

Table 2: Selected pairwise Pearson correlations ($|\rho| > 0.44$).

Feature pair	ρ	Interpretation
<code>sqft_living</code> – <code>grade</code>	0.763	Size and quality linked
<code>sqft_living</code> – <code>sqft_living15</code>	0.756	Neighbourhood size effect
<code>sqft_living</code> – <code>bathrooms</code>	0.755	More space, more amenities
<code>grade</code> – <code>bathrooms</code>	0.665	Quality and amenity count
<code>bathrooms</code> – <code>bedrooms</code>	0.529	Room count correlated
<code>floors</code> – <code>house_age</code>	-0.489	Newer homes multi-floor
<code>grade</code> – <code>house_age</code>	-0.447	Newer builds higher grade

KNIME Pipeline Architecture

The full preprocessing pipeline in KNIME consists of the following nodes in sequence: CSV Reader → Row Filter (outlier removal) → Column Filter 1 (remove `id`, `date`, `zipcode`) → Math Formula (`house_age`) → Rule Engine ×2 (`renovated`, `basement`) → Column Filter 2 (remove `yr_built`, `yr_renovated`, `sqft_basement`, `sqft_above`) → Missing Value → Math Formula (`log_price`) → Linear Correlation → Correlation Filter → Table Partitioner → Normalizer / Normalizer Apply. From the normalised outputs, five parallel branches implement the regression models, PCA, cross-validation loops, and clustering.

Models and Methods

Regression Models

Four models of increasing complexity are trained to predict `log_price` from the 14 normalised features.

Linear Regression (LR)

Ordinary least squares regression serves as the interpretable baseline. LR assumes a linear relationship between features and log-price. While this assumption is unlikely to hold perfectly, LR provides a competitive lower bound and a set of interpretable coefficients. The target column is set to `log_price`; the feature set includes all 14 variables with `price` excluded.

Decision Tree (DT)

The CART (Classification and Regression Trees) algorithm is used with maximum depth=8 and minimum 5 records per node. This configuration captures non-linear relationships and feature interactions without excessive overfitting. Depth=8 represents a deliberate balance between model expressiveness and generalisation ability.

Random Forest (RF)

An ensemble of 100 decorrelated regression trees (seed=42, square-root feature sampling per split). Each tree is trained on a bootstrap sample of the training data; predictions are averaged across all trees. This bagging strategy reduces variance substantially without increasing bias, making RF robust to outliers and less prone to overfitting than a single DT.

Gradient Boosted Trees (GBT)

A sequential ensemble of 200 trees (learning rate=0.05, max depth=6, row sampling=0.8). Each tree corrects the residual errors of the current ensemble. The small learning rate (0.05) promotes slow, stable convergence and reduces the risk of overfitting. Row sampling (0.8) introduces additional stochasticity that further regularises the model.

Dimensionality Reduction via PCA

Principal Component Analysis is applied to understand the structure of the feature space and to quantify how much predictive information is retained when dimension-

ality is reduced. PCA Compute is applied to all 14 normalised training features (`price` and `log_price` excluded, as they are the regression targets). The number of components is determined by the criterion of retaining 95% of total variance, yielding **12 principal components**. PCA Apply then projects both training and test sets using the identical learned transformation, maintaining the train/test separation.

A new Linear Regression model is trained exclusively on the 12-dimensional projected training set and evaluated on the projected test set. This controlled comparison isolates the effect of dimensionality reduction: any performance difference between this LR+PCA model and the full LR model is attributable solely to the information lost in compression.

5-Fold Cross-Validation

Cross-validation provides a more reliable estimate of generalisation performance than a single holdout split, because it averages results over multiple different train/test partitions. Independent 5-fold CV loops are implemented for each of the four regression models using separate X-Partitioner/X-Aggregator node pairs. Each partitioner receives the full preprocessed dataset before the holdout split, with linear sampling and seed=42 to ensure reproducibility.

Using separate partitioners per model — rather than a single shared partitioner with all models inside the same loop — is a deliberate KNIME design choice that avoids loop-conflict errors and ensures fully independent evaluation for each model. The X-Aggregator collects fold-level predictions and computes iteration statistics across all five folds.

K-Means Clustering

K-Means clustering is applied as an unsupervised complement to the regression analysis, with the goal of identifying natural market segments in the dataset. A dedicated feature subset of 10 variables is selected for clustering: `sqft_living`, `grade`, `lat`, `long`, `price`, `bathrooms`, `view`, `waterfront`, `house_age`, `renovated`. Geographic features (`lat`, `long`) are included to capture spatial price patterns. Z-score normalisation is applied to the full dataset (not only the training split) before clustering, as clustering is an exploratory analysis not subject to the train/test protocol.

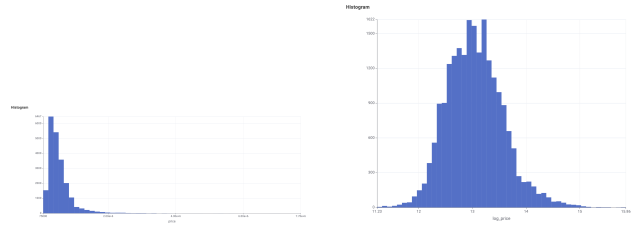
Values of $K \in \{2, 3, 4, 5\}$ are systematically evaluated using the Silhouette Coefficient [6], which measures cohesion within clusters and separation between them. The optimal K is selected as the value maximising the overall Silhouette score. Results are visualised using a Color Manager node (cluster assignment) connected to a Scatter Plot ($X=\text{long}$, $Y=\text{lat}$) and a GroupBy node computing cluster-level statistics (mean price, living space, grade, bathroom count).

Results and Discussion

Price Distribution and Log Transformation

Figure 1 demonstrates the effect of log transformation. The raw price distribution (left) is strongly right-skewed with a long upper tail driven by a small number of luxury properties. After log transformation (right), the distri-

bution is approximately Gaussian and symmetric. This validates our preprocessing choice and confirms that the log-price scale is appropriate for regression modelling.



(a) Raw price: right-skewed. (b) Log-price: approx. normal.

Figure 1: Effect of log transformation on the target distribution. The normalised shape satisfies linear regression assumptions.

Regression Performance

Table 3 reports the full holdout and cross-validation results for all five models. All metrics are computed on the log-price scale.

Table 3: Holdout and 5-fold CV results (target: `log_price`). Best values in bold.

Model	R^2	RMSE	MAE	CV MSE	CV RMSE
Random Forest	0.703	0.292	0.231	0.082	0.286
GBT	0.696	0.295	0.231	0.087	0.296
Linear Regression	0.663	0.311	0.247	0.095	0.308
LR + PCA (12 PCs)	0.654	0.315	0.229	—	—
Decision Tree	0.620	0.330	0.260	0.107	0.327

Random Forest

RF achieves the best performance on both holdout ($R^2=0.703$) and cross-validation (CV RMSE=0.286), confirming that the variance reduction from bagging over 100 trees translates to superior generalisation on unseen data. The close agreement between holdout and CV results indicates that the model is stable and not over-fitted to the specific train/test split.

Gradient Boosted Trees

GBT ranks second overall ($R^2=0.696$, CV RMSE=0.296). Its CV RMSE is only 0.010 higher than RF, suggesting competitive performance. The sequential boosting strategy effectively exploits the residuals from earlier trees, but the marginal gain over RF is modest on this dataset. Importantly, GBT’s holdout and CV rankings are consistent, confirming good generalisation.

Linear Regression

LR is competitive ($R^2=0.663$), demonstrating that log-transformed price has a substantially linear relationship with the 14 engineered features. The strong baseline performance of LR — only 0.040 R^2 below RF — confirms that the preprocessing choices (log transform, feature engineering) contribute significantly to model quality across all algorithms.

LR + PCA

The LR model trained on 12 principal components achieves $R^2=0.654$, compared to $R^2=0.663$ for full LR on all 14 features. The difference of 0.009 R^2 units quantifies the information cost of compressing 14 features into 12 orthogonal PCs. This small gap has two important implications: (i) the 14 features contain limited redundancy — 12 PCs are sufficient to capture nearly all predictive signal; (ii) PCA is a viable dimensionality reduction strat-

egy for this problem when compression is required. The slight improvement in MAE for LR+PCA (0.229 vs. 0.231 for RF) is attributable to the orthogonalisation of the feature space by PCA, which may reduce the influence of correlated features on individual coefficient estimates.

Decision Tree

DT achieves $R^2=0.620$ on holdout but performs worst in cross-validation (CV RMSE=0.327). The gap between holdout and CV performance — the largest among all models — is a clear sign of overfitting: the tree memorises patterns specific to the training split that do not generalise. This confirms that depth=8, while capturing useful non-linearities, allows the tree to fit noise as well as signal.

Residual Analysis

Figure 2 presents the residual diagnostics for Random Forest on the holdout test set. The residual histogram (left) shows a bell-shaped, symmetric distribution tightly centred at zero. The mean signed difference is -0.00025 , confirming that the model’s predictions are essentially unbiased. The scatter plot of residuals versus predicted values (right) shows no systematic pattern, fan-shape, or curvature — confirming homoscedasticity (constant error variance) across the prediction range. Both diagnostics jointly validate the modelling assumptions and confirm that RF produces well-calibrated predictions.

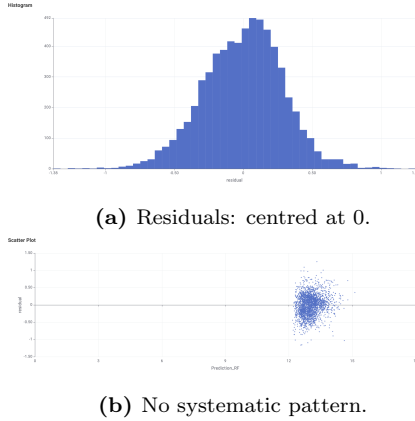


Figure 2: Residual analysis for Random Forest (holdout test set). Both diagnostics confirm unbiased, homoscedastic errors, validating regression assumptions.

K-Means Clustering

Cluster Selection

Table 4 reports Silhouette scores for $K = 2, 3, 4, 5$. The scores decrease monotonically as K increases. Of particular note, $K = 3$ already introduces a cluster with a negative per-cluster Silhouette score (-0.053 for cluster_2), indicating that this partition forces artificial separation on points that are better described by the $K = 2$ structure. $K = 2$ is therefore selected as the optimal number of clusters.

Table 4: Overall Silhouette scores for $K = 2, 3, 4, 5$.

K	Overall Silhouette	Min cluster score
2	0.248	0.103
3	0.217	-0.053
4	0.199	-0.061
5	0.182	-0.070

Cluster Characteristics

With $K=2$, two economically and geographically distinct market segments emerge:

Cluster 0 — Standard market: 13,961 properties (65% of the dataset). This cluster is characterised by below-average normalised price (-0.39 SD), below-average living space (-0.51 SD), lower construction grade (-0.52 SD), and older properties ($+0.33$ SD house age). It represents the broad, accessible segment of the King County housing market.

Cluster 1 — Premium market: 7,651 properties (35% of the dataset). This cluster shows significantly higher normalised price ($+0.70$ SD), larger living area ($+0.94$ SD), higher construction grade ($+0.95$ SD), and a substantially greater proportion of waterfront and renovated properties. It represents the premium, high-value segment.

Geographic Distribution

Figure 3 shows the geographic distribution of clusters on normalised latitude/longitude coordinates. The premium cluster (blue) concentrates in northern and eastern areas of King County, while the standard cluster (orange) is more broadly distributed. This spatial separation confirms that location is a key driver of cluster membership, consistent with the inclusion of `lat` and `long` among the clustering features. The geographic coherence of the clusters adds credibility to the segmentation result: the two groups are not statistical artefacts but reflect real estate market geography.



Figure 3: K-Means ($K=2$) geographic distribution. Orange = standard market (Cluster 0, 65%); blue = premium market (Cluster 1, 35%). Spatial separation reflects real geographic price gradients in King County.

Feature Importance

RF attribute statistics (number of splits at tree level 0, i.e. the root of each tree) identify the most globally discriminative features: `bathrooms` (20 root splits), `sqft_living15` (19), `sqft_living` (16), `grade` (13), and `floors` (11). These findings are consistent with the correlation analysis in Table 2: features most correlated with `log_price` are also those used most frequently at the root of trees, where splits have the largest impact on prediction error.

Conversely, `condition`, `waterfront`, and `renovated` show zero root-level splits, indicating that while these features carry predictive signal at deeper levels of the trees, they are not globally dominant price drivers on their own. This validates the feature engineering choices: these binary features contribute to the model through interactions with continuous features rather than as pri-

mary standalone predictors. The consistency between correlation analysis, regression coefficients, tree-level feature importance, and clustering statistics — all pointing to living space, grade, and bathroom count as the primary price drivers — reinforces the robustness of this finding.

Discussion and Conclusions

Model Performance Summary

This study demonstrates that machine learning models can predict King County house prices with high accuracy from structural and locational features. Random Forest achieves the best performance ($R^2=0.703$, $RMSE=0.292$), a result that is stable and reproducible across holdout evaluation and 5-fold cross-validation ($CV\ RMSE=0.286$). GBT is a strong competitive alternative that may be preferred in scenarios requiring sequential error correction. LR provides a robust and interpretable baseline, confirming that the engineered features have a substantially linear relationship with log-price.

The DT result highlights an important practical lesson: models that perform well on a single holdout split may overfit. The large gap between DT holdout performance ($R^2=0.620$) and CV performance ($RMSE=0.327$) — the widest among all models — underscores the importance of cross-validation for reliable model selection.

PCA Findings

PCA with 12 components (95% variance) achieves $R^2=0.654$, only 0.009 below full LR (0.663). This result has a precise interpretation: the 14 preprocessed features contain limited redundancy — once the most correlated groups are accounted for by PCA, very little additional variance is captured by the remaining components. PCA is therefore a viable option for this problem when interpretability or computational constraints motivate dimensionality reduction. The result also confirms that the feature engineering decisions (log transform, derived features, removal of redundant columns) produced a clean, informative feature set with minimal noise.

Clustering Insights

K-Means with $K=2$ (Silhouette=0.248) reveals a premium/standard market split that is not only statistically valid but also geographically coherent and economically meaningful. The 35%/65% split between premium and standard segments, the large differences in living space (+0.94 SD) and grade (+0.95 SD), and the geographic concentration of premium properties in northern King County all align with known Seattle-area real estate patterns. This segmentation has direct practical value for pricing strategy, targeted marketing, and investment analysis.

Limitations and Future Work

The dataset covers a single year (2014–2015) and one geographic region, limiting temporal and spatial generalisability. Models do not incorporate macroeconomic variables (interest rates, housing supply) or spatial autocorrelation, which are known price drivers. The K-Means clustering result uses Euclidean distance in normalised space, which may not fully capture the true price geography of the region; density-based methods such as DBSCAN could reveal non-spherical cluster structures.

Future work could extend the pipeline with spatial regression models, time-series components capturing market dynamics, and more powerful ensemble methods. An XGBoost or LightGBM implementation could potentially improve on the GBT $R^2=0.696$ result. Expanding the feature set with school district ratings, crime indices, or proximity to urban centres could also improve model quality.

Reproducibility. The full KNIME workflow, dataset, and all result plots are publicly available:

<https://github.com/aminentezari/>



Machine-learning-house-price

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